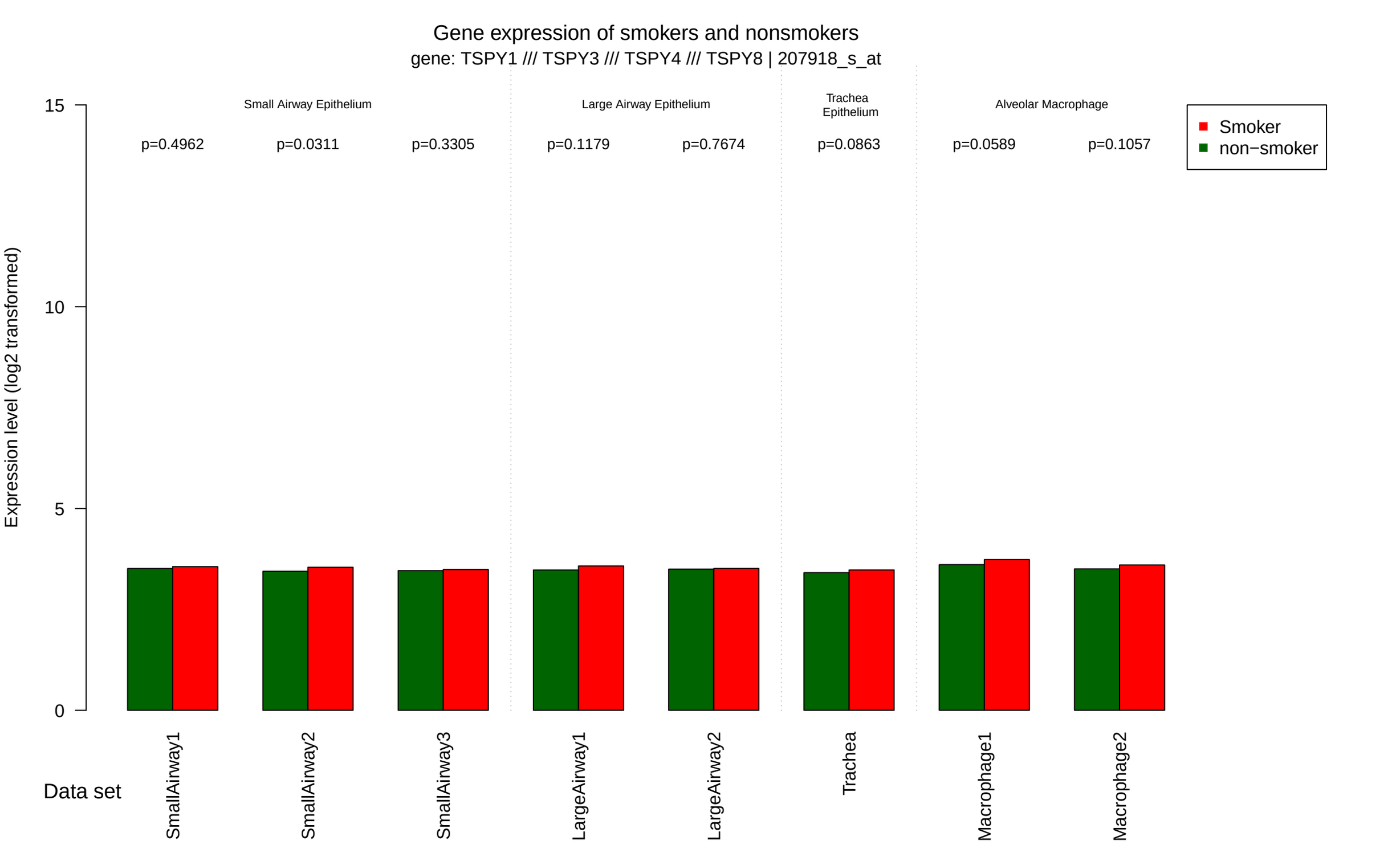
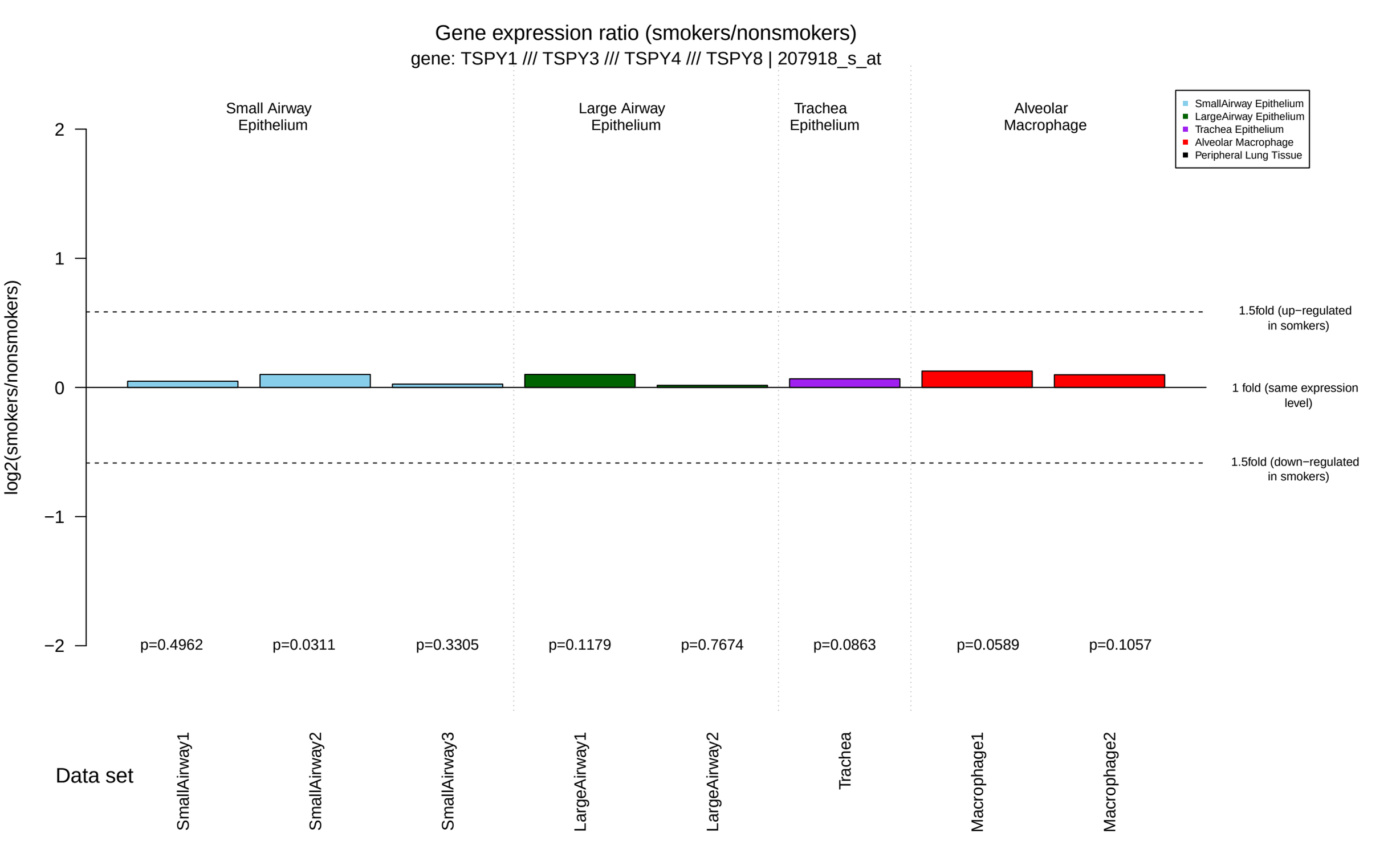
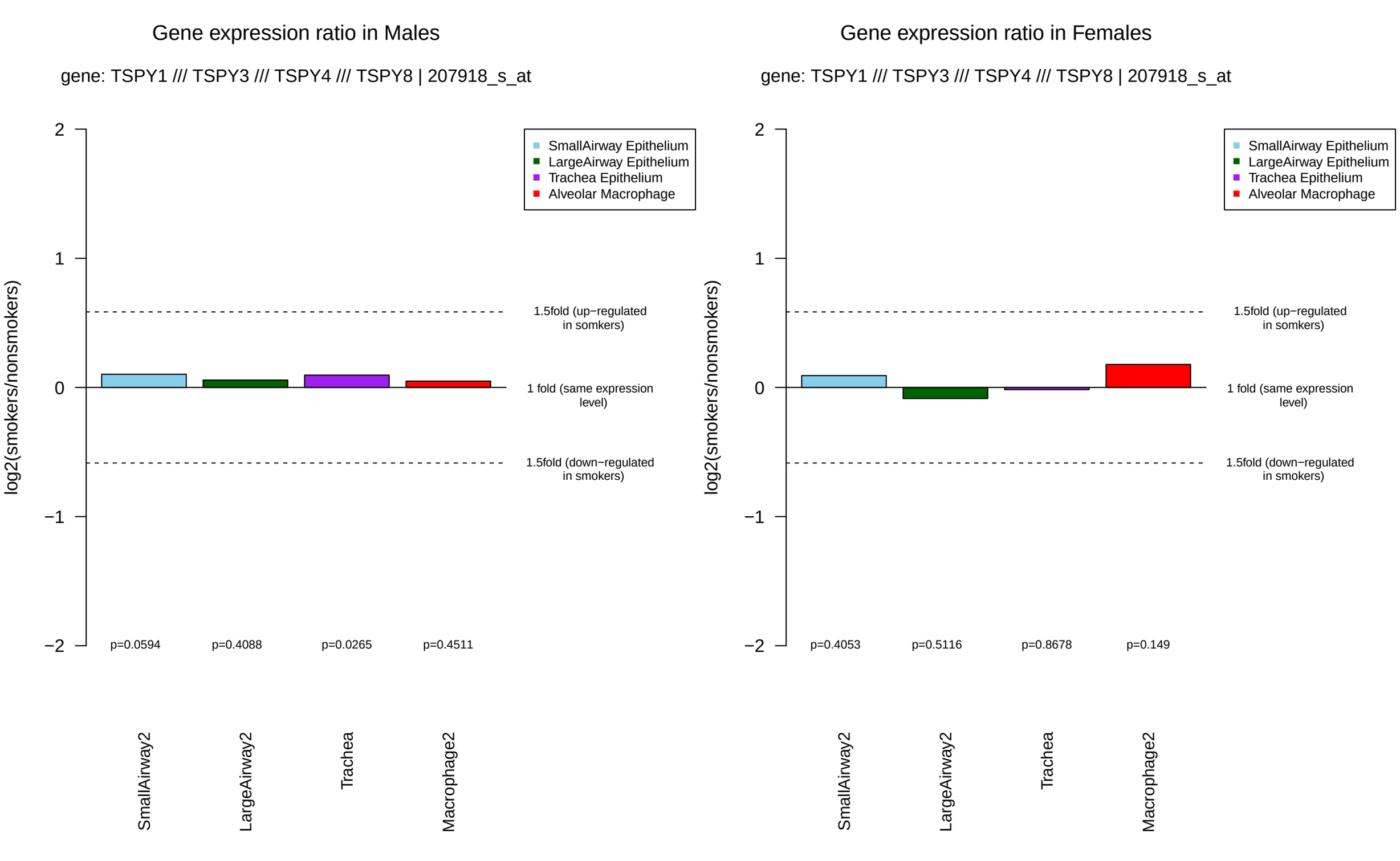
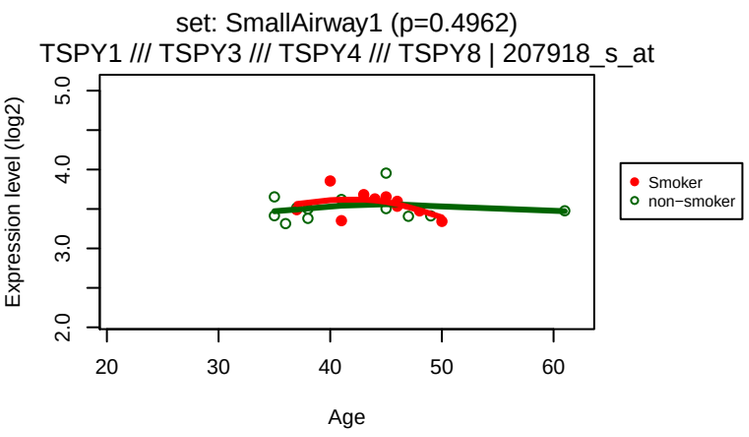
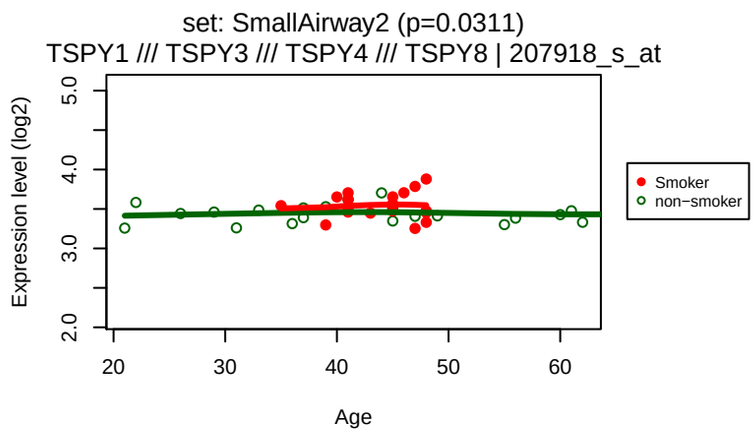
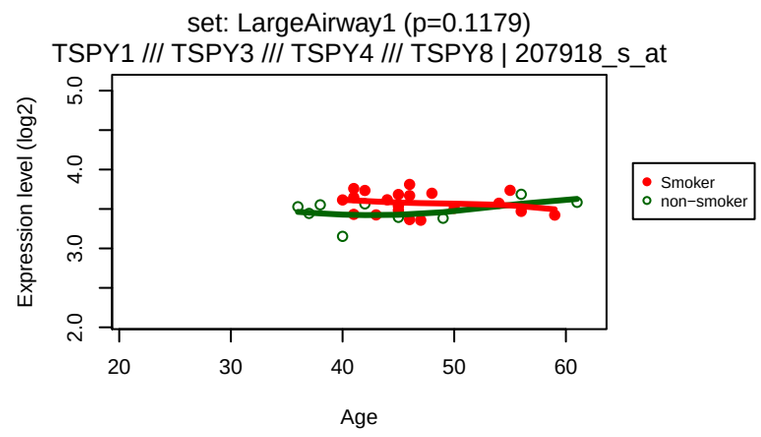

Gene expression of smokers and nonsmokers
gene: TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4962
p=0.0311
p=0.3305
p=0.1179
p=0.7674
p=0.0863
p=0.0589
p=0.1057
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

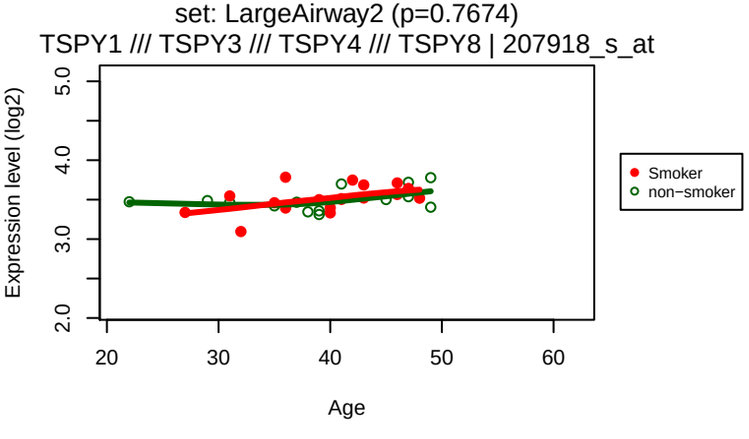

Gene expression ratio (smokers/nonsmokers)
gene: TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4962
p=0.0311
p=0.3305
p=0.1179
p=0.7674
p=0.0863
p=0.0589
p=0.1057
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

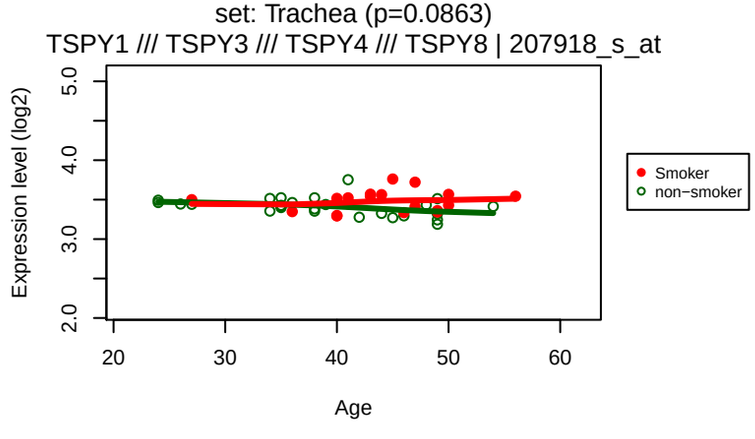

Gene expression ratio in Males
gene: TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0594
p=0.4088
p=0.0265
p=0.4511
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4053
p=0.5116
p=0.8678
p=0.149
SmallAirway2
LargeAirway2
Trachea
Macrophage2

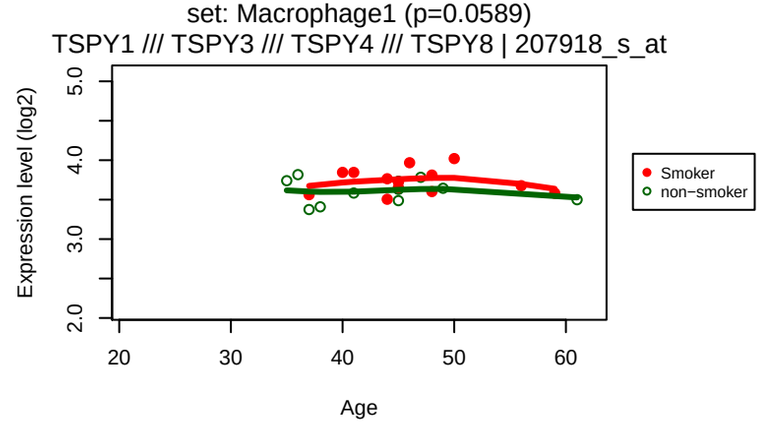

set: SmallAirway1 (p=0.4962)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker

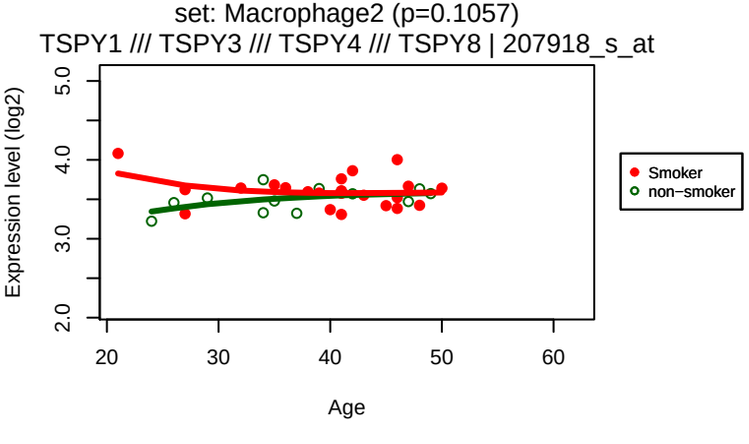

set: SmallAirway2 (p=0.0311)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker

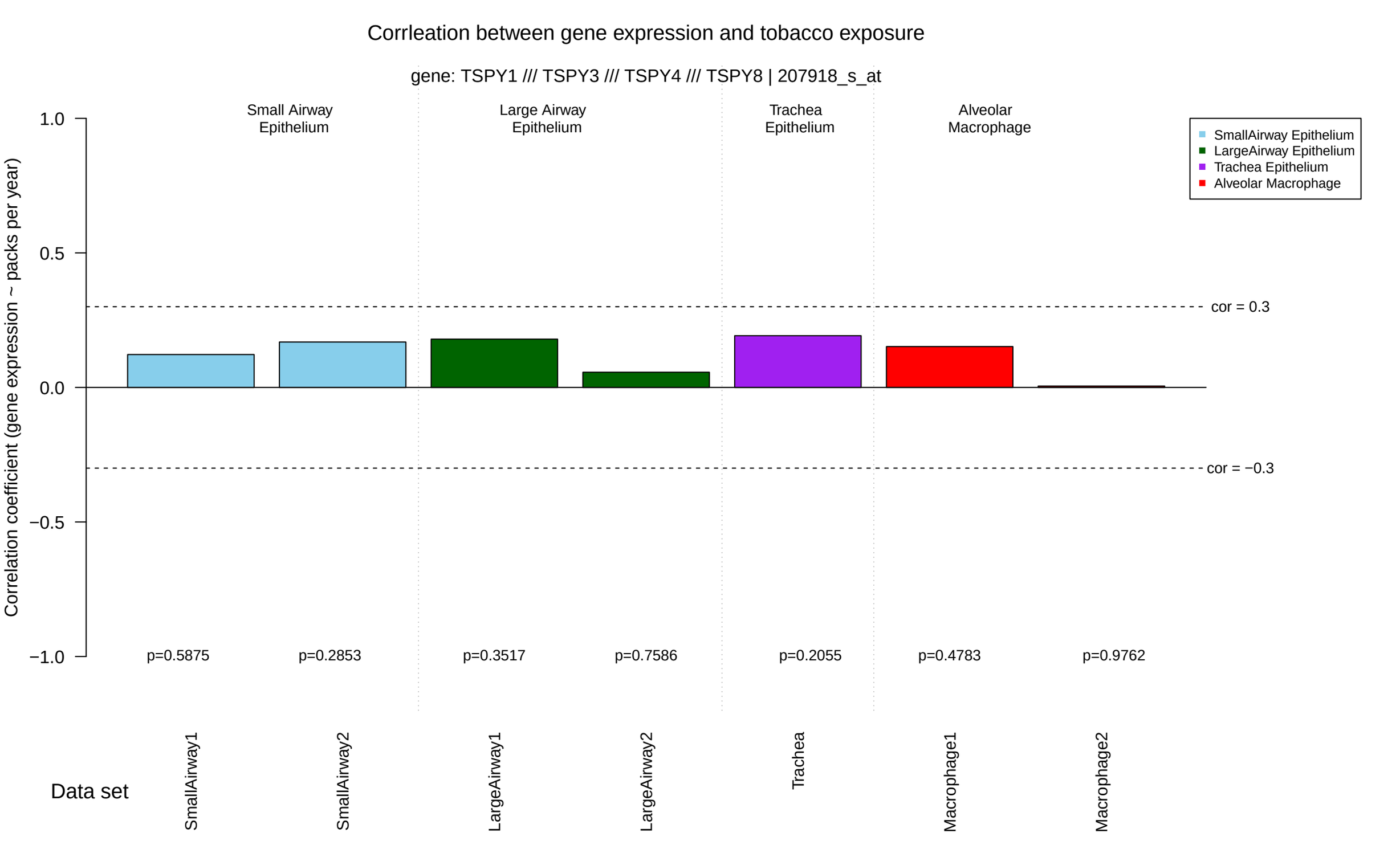

set: LargeAirway1 (p=0.1179)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7674)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0863)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0589)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1057)
TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPY1 /// TSPY3 /// TSPY4 /// TSPY8 | 207918_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5875
p=0.2853
p=0.3517
p=0.7586
p=0.2055
p=0.4783
p=0.9762
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2